

DW_8b10b_unbal

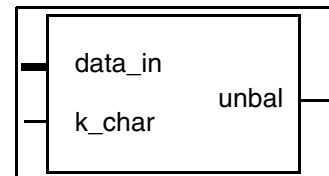
8b10b Coding Balance Predictor

Version, STAR and Download Information: [IP Directory](#)

Features and Benefits

- Independent of Running Disparity
- Higher speed than a full encoder
- Predicts balance for both data and special characters

Revision History



Description

DW_8b10b_unbal predicts (without fully encoding) whether a given character will or will not flip the running disparity of the 8b/10b encoder when it is encoded. This function is useful for implementing multi-byte encoder schemes with some protocols that tightly control the state of the running disparity in the encoder. DW_8b10b_unbal may also be useful even in single byte encoder designs that require pipelining as well as control of the running disparity.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
k_char	1 bit	Input	Special character control input (low for data characters, high for special characters)
data_in	8 bits	Input	Input for 8-bit data character to be encoded
unbal	1 bit	Output	Unbalanced code character indicator (low for balanced, high for unbalanced)

Table 1-2 Parameter Description

Parameter	Values	Description
k28_5_mode	0 or 1 Default: 0	Special Character subset control parameter ■ 0 for all special characters available ■ 1 for only K28.5 available (when k_char = high, regardless of the value on data_in)

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	DesignWare

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```
- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN
```

 (which is used for the Synopsys VCS simulator)

The warning messages for this model include the following:

- If data_in contains invalid data for a specific configuration, the following message is displayed:

```
WARNING: <instance_path>:  
    at time = <timestamp>, Invalid data on data_in of DW_8b10b_unbal when k28_5_only=0  
    and k_char=1.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Related Topics

- [Coding Group – Coding Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_foundation_comp.all;

entity DW_8b10b_unbal_inst is
  generic (inst_k28_5_only : integer := 0 );
  port (inst_k_char   : in std_logic;
        inst_data_in  : in std_logic_vector(7 downto 0);
        unbal_inst    : out std_logic );
end DW_8b10b_unbal_inst;

architecture inst of DW_8b10b_unbal_inst is
begin

  -- Instance of DW_8b10b_unbal
  U1 : DW_8b10b_unbal
    generic map (k28_5_only => inst_k28_5_only )
    port map (k_char => inst_k_char,   data_in => inst_data_in,
              unbal => unbal_inst );
end inst;

-- Configuration for use with VSS simulator
-- pragma translate_off
configuration DW_8b10b_unbal_inst_cfg_inst of DW_8b10b_unbal_inst is
  for inst
    end for; -- inst
end DW_8b10b_unbal_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_8b10b_unbal_inst( inst_k_char, inst_data_in, unbal_inst );

    parameter k28_5_only = 0;

    input inst_k_char;
    input [7 : 0] inst_data_in;
    output unbal_inst;

    // Instance of DW_8b10b_unbal
    DW_8b10b_unbal #(k28_5_only)
        U1 (.k_char(inst_k_char), .data_in(inst_data_in), .unbal(unbal_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Added “Suppressing Warning Messages During Verilog Simulation” on page 2
March 2019	DWBB_201903.0	■ Removed minPower designation from this datasheet
January 2019	DWBB_201806.5	■ Updated example in “HDL Usage Through Component Instantiation - VHDL” on page 3 ■ Added this Revision History table and the document links on this page

Copyright Notice and Proprietary Information

© 2024 Synopsys, Inc. All rights reserved. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>.

All other product or company names may be trademarks of their respective owners.

Free and Open-Source Software Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

Synopsys, Inc.
www.synopsys.com